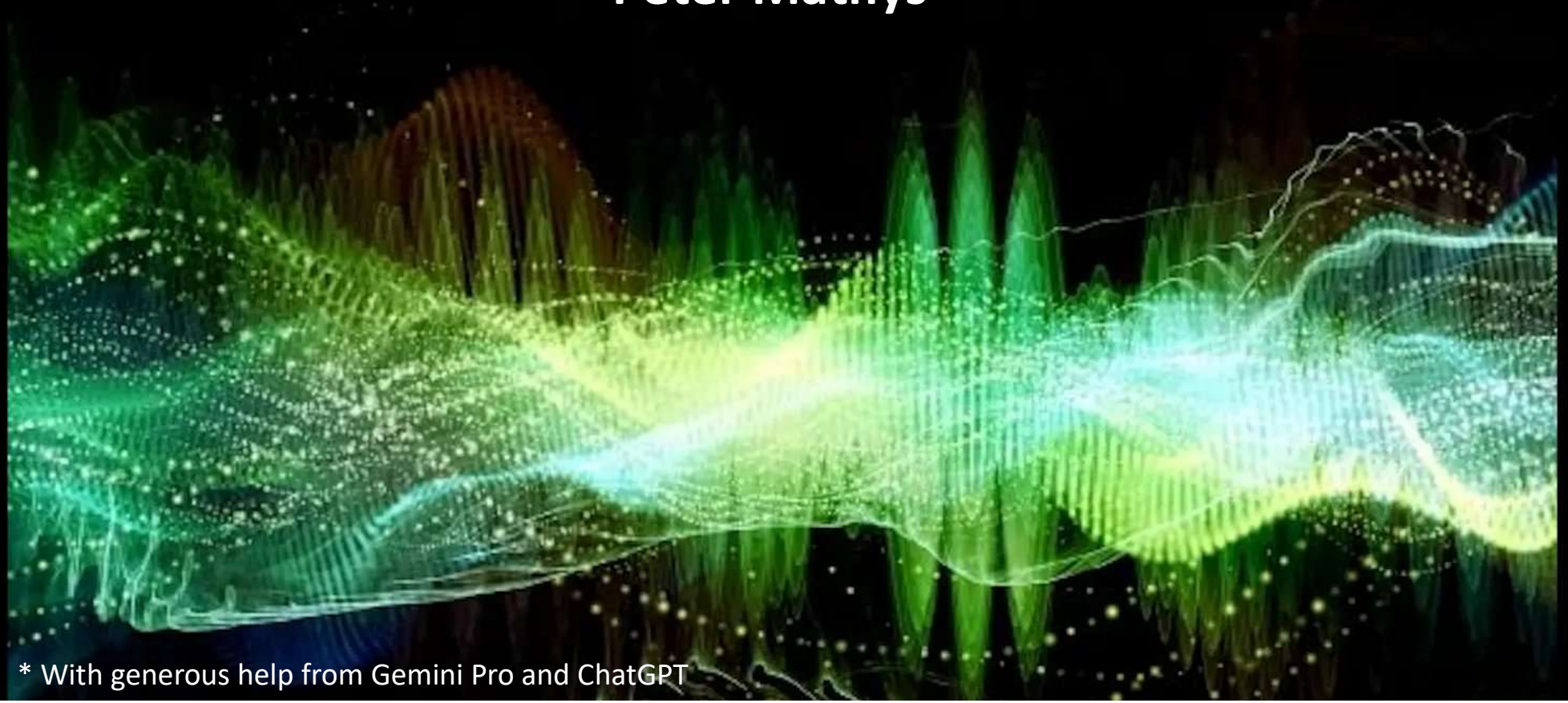
ECEN 4/5632 Theory and Application of Digital Filters

Lecture 25, 11-10-26

Peter Mathys*



* With generous help from Gemini Pro and ChatGPT

Feature Engineering

From Physics to Perception

STFT → Mel filterbanks → log compression → MFCCs → fixed-size ML inputs

WEEK 13 • LECTURE 25 • 75 MINUTES

Week 13 changes the objective

THE TRANSITION FROM DSP ANALYSIS TO MACHINE LEARNING

00–15 MIN

- Classical DSP asks: can we represent, filter, or reconstruct the signal faithfully?
- ML asks: which representation makes the decision boundary easiest to learn?
- A feature may intentionally discard information that is irrelevant to the task.
- But discarded information cannot be recovered later—feature engineering encodes assumptions.

Preserve what matters for the label; compress what does not.

Feature engineering = physics + perception + task knowledge

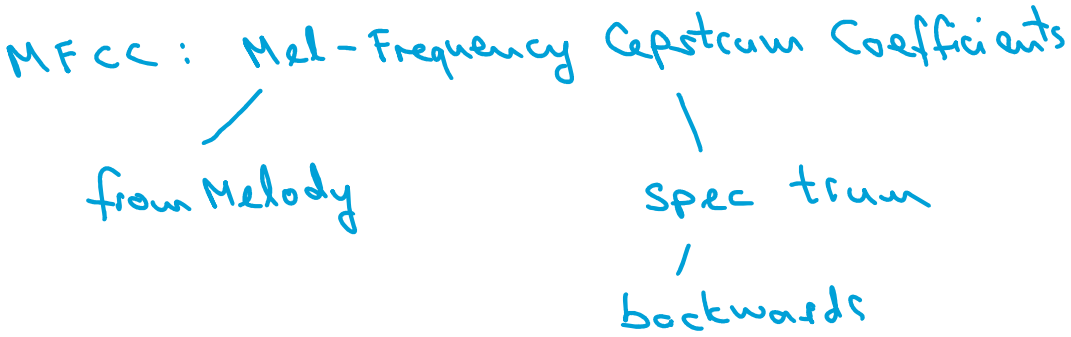
Instructor cue: Ask for one task where perfect reconstruction is unnecessary: siren detection, keyword spotting, bearing-fault classification.

Learning outcomes

BY THE END OF LECTURE 21

00–15 MIN

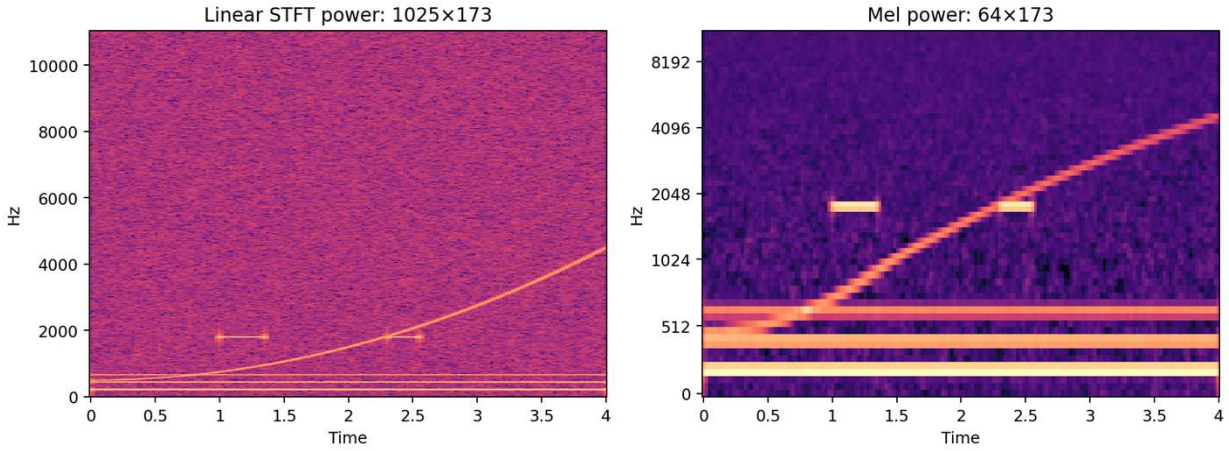
- Explain why linear-Hz STFT bins are not always the most efficient coordinates for audio classification.
- Construct a Mel-spectrogram conceptually as a weighted sum of STFT power bins.
- Explain log compression and the DCT step used to form MFCCs.
- Standardize waveform geometry: channels, sample rate, duration, and feature shape.
- Choose between Mel-spectrograms, MFCC matrices, and summarized MFCC vectors for different model families.



The problem with linear frequency

THE STFT IS PHYSICALLY NATURAL, NOT PERCEPTUALLY UNIFORM

00–15 MIN



- At 22.05 kHz with n_fft=2048, the one-sided STFT has 1025 frequency bins.
- For many audio tasks, fine resolution at the top of the band is less useful than fine resolution at low frequencies.
- Mel filterbanks pool neighboring bins nonuniformly, reducing dimension while keeping time structure.

$$\text{STFT power } P[k,m] \rightarrow \text{Mel power } S_{\text{mel}}[b,m]$$

Instructor cue: Phrase "human hearing is logarithmic" carefully: Mel is a perceptual warping, not a universal law of hearing.

Physics coordinates vs perception-oriented coordinates

SAME SOUND, DIFFERENT REPRESENTATIONS

00–15 MIN

Linear Hz / STFT
Equal-width frequency bins
Direct physical meaning
Filter-analysis friendly

Mel
More low-frequency detail
Broader high-frequency bands
Compact audio front end

MFCC
DCT of log-Mel
Compact spectral envelope
Classical baseline friendly

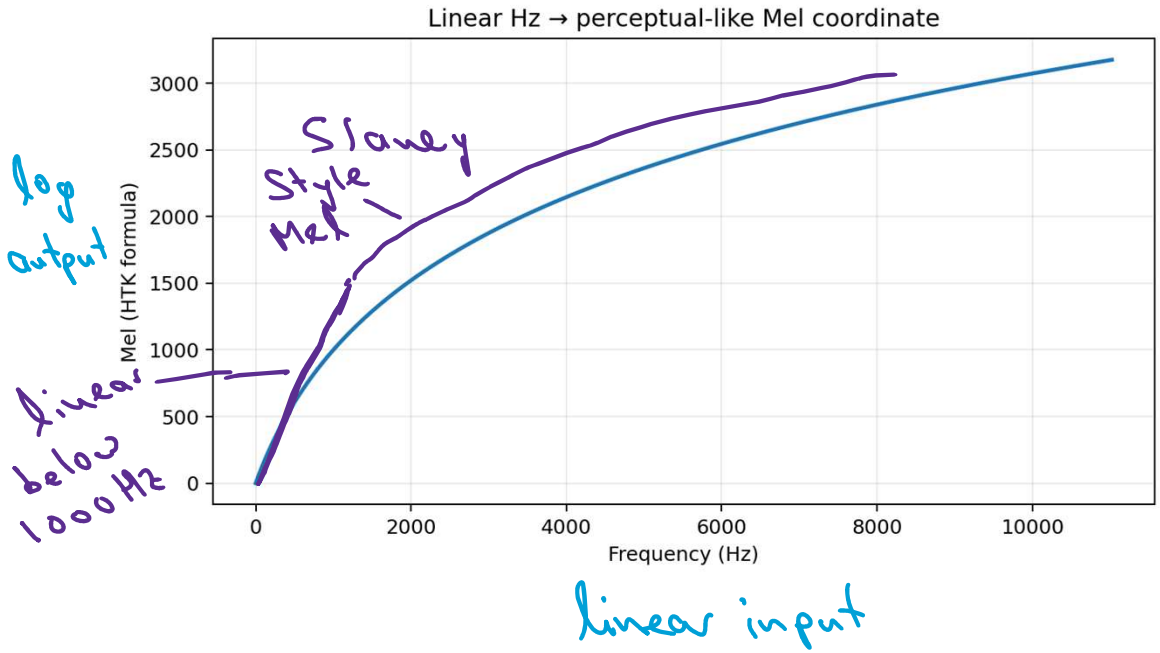
CNN input
Keep full 2D map
Learn local patterns
Next week: CNNs

No representation is free: every compression trades detail for invariance or efficiency.

The Mel scale

ONE COMMON MAPPING FROM HZ TO A PERCEPTUAL-LIKE COORDINATE

15–35 MIN



$$m = 2595 \log_{10}(1 + f/700)$$

- The formula shown is the common HTK-style mapping.
- librosa defaults to a Slaney-style Mel definition; set htk=True for the HTK convention.
- The important idea is frequency warping—not memorizing one constant.

HTK: Hidden Markov Toolkit (Cambridge speech recognition toolkit

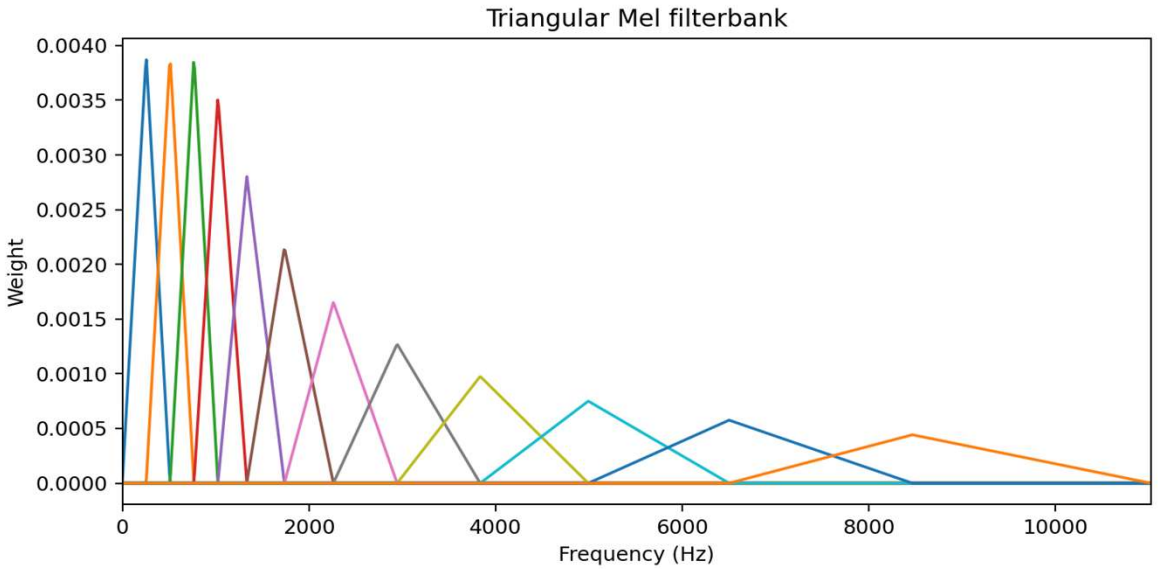
torchaudio uses HTK as default

Instructor cue: Use 100→200 Hz versus 10,000→10,100 Hz as intuition; avoid claiming an exact perceptual threshold.

Mel filterbanks are weighted pooling

TRIANGULAR BANDS COMPRESS FREQUENCY WITHOUT DESTROYING TIME

15–35 MIN



- Each filter H_b[k] assigns weights to neighboring FFT bins.
- Low-frequency filters are relatively narrow; high-frequency filters cover more Hz.
- Overlapping filters provide smooth transitions between bands.

$$S_mel[b,m] = \Sigma_k \, H_b[k] \, |X[m,k]|^2$$

STFT → Mel freq conversion

Matrix view: Mel features are a linear map of power

A BRIDGE BACK TO LINEAR ALGEBRA

15–35 MIN



- Frequency compression happens independently in every time frame.
- Time resolution is inherited from the STFT hop length.
- Choosing n_mels controls the frequency-feature dimension.

librosa: compute a Mel-spectrogram

CODE SHOULD MIRROR THE MATH

15–35 MIN

n_fft = 2048
1025 linear spaced frequencies

n_mels = 64
mel-bands

```
M = librosa.feature.melspectrogram(
        y=y, sr=22050,
    n_fft=2048, hop_length=512,
            n_mels=64,
            power=2.0,
              )
M_db = librosa.power_to_db(M, ref=np.max)

  print(M.shape)      # (64, time_frames)
print(M_db.max())   # 0 dB relative to peak
```

reference is 0 dB

$10 \log_{10}(S_{mel}[m,t])$

Instructor cue: Ask students to predict the first dimension. power=2 means power; power_to_db therefore uses 10 log10.

Why take the logarithm?

DYNAMIC RANGE COMPRESSION IS PART OF FEATURE DESIGN

15–35 MIN

$$S_{dB} = 10 \log_{10}(S / S_{ref})$$

- Spectral power can span orders of magnitude.
- Log compression turns ratios into differences and makes weak structure more visible.
- For power quantities use $10 \log_{10}(\cdot)$; for amplitude ratios use $20 \log_{10}(\cdot)$.
- A floor is required numerically because log(0) is undefined.

100× power ratio → +20 dB

0.01× power ratio → −20 dB

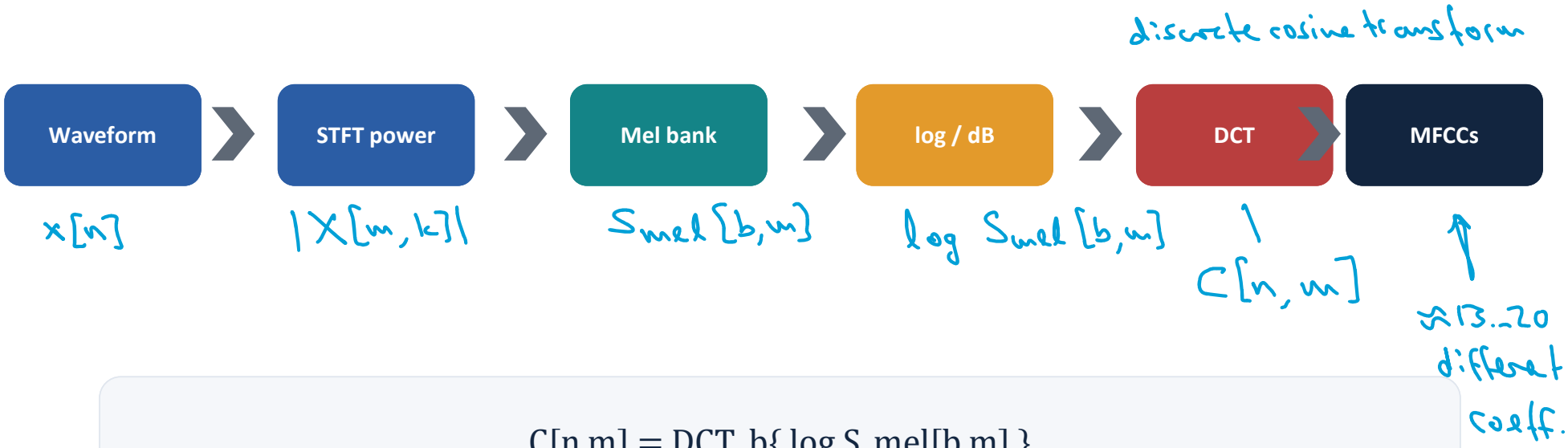
large dynamic range is captured better by log

floor: $10 \log_{10}(S_{mel}[m, b] + \varepsilon)$ $\varepsilon \approx 1e-12$

MFCC pipeline

A COMPACT DESCRIPTION OF SPECTRAL-ENVELOPE SHAPE

35–55 MIN



$$C[n,m] = \mathrm{DCT}_b\{ \log S_{\mathrm{mel}}[b,m] \}$$

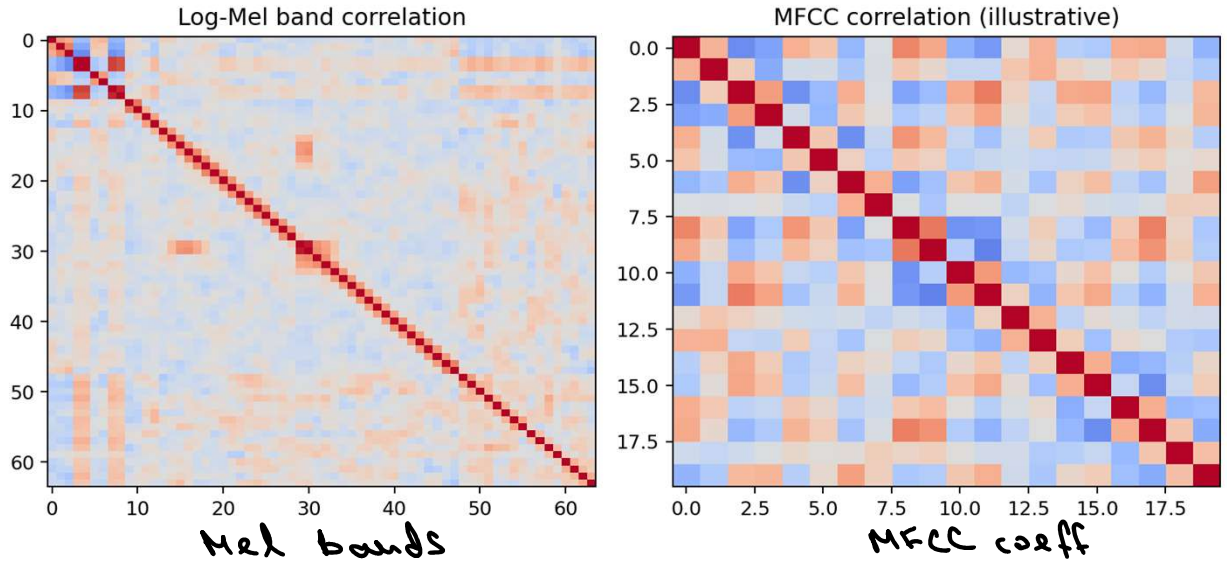
- Many classical pipelines retain roughly the first 13–40 coefficients.
- Coefficient 0 largely tracks overall log-energy; higher indices encode faster variation across Mel bands.

What does the DCT buy us?

APPROXIMATE DECORRELATION AND ENERGY COMPACTION

35–55 MIN

MFCC



- Neighboring Mel bands overlap, so their log energies are correlated.
- The DCT rotates those coordinates into cosine-basis directions.
- Low-order coefficients often concentrate useful smooth spectral-envelope information.
- "Decorrelates" is approximate and data-dependent—not a universal guarantee.

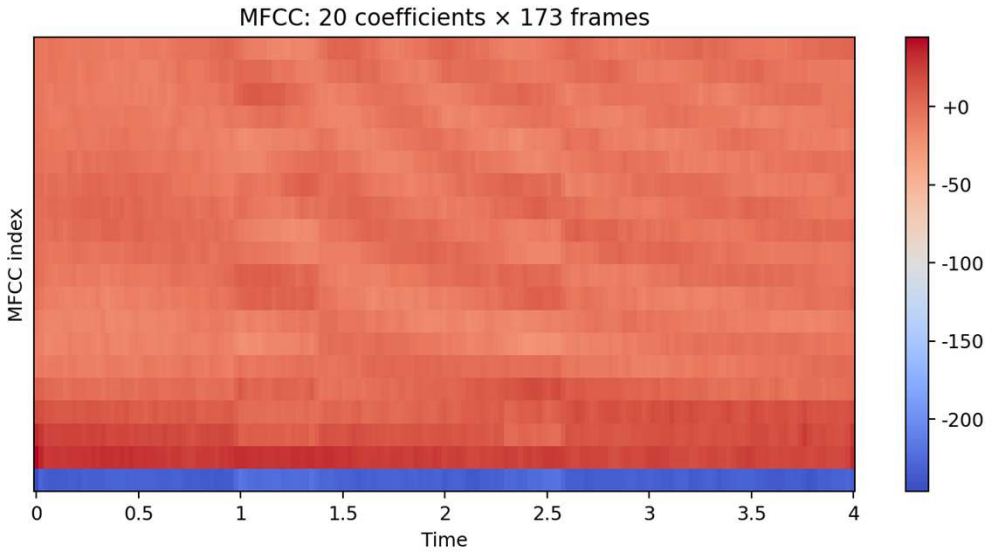
DCT helped produce compact coordinates that are mildly decorrelated and that are easier for classical statistical models to handle

Instructor cue: Connect to older Gaussian/diagonal-covariance models. The DCT is not magic statistical independence.

How should students interpret MFCC indices?

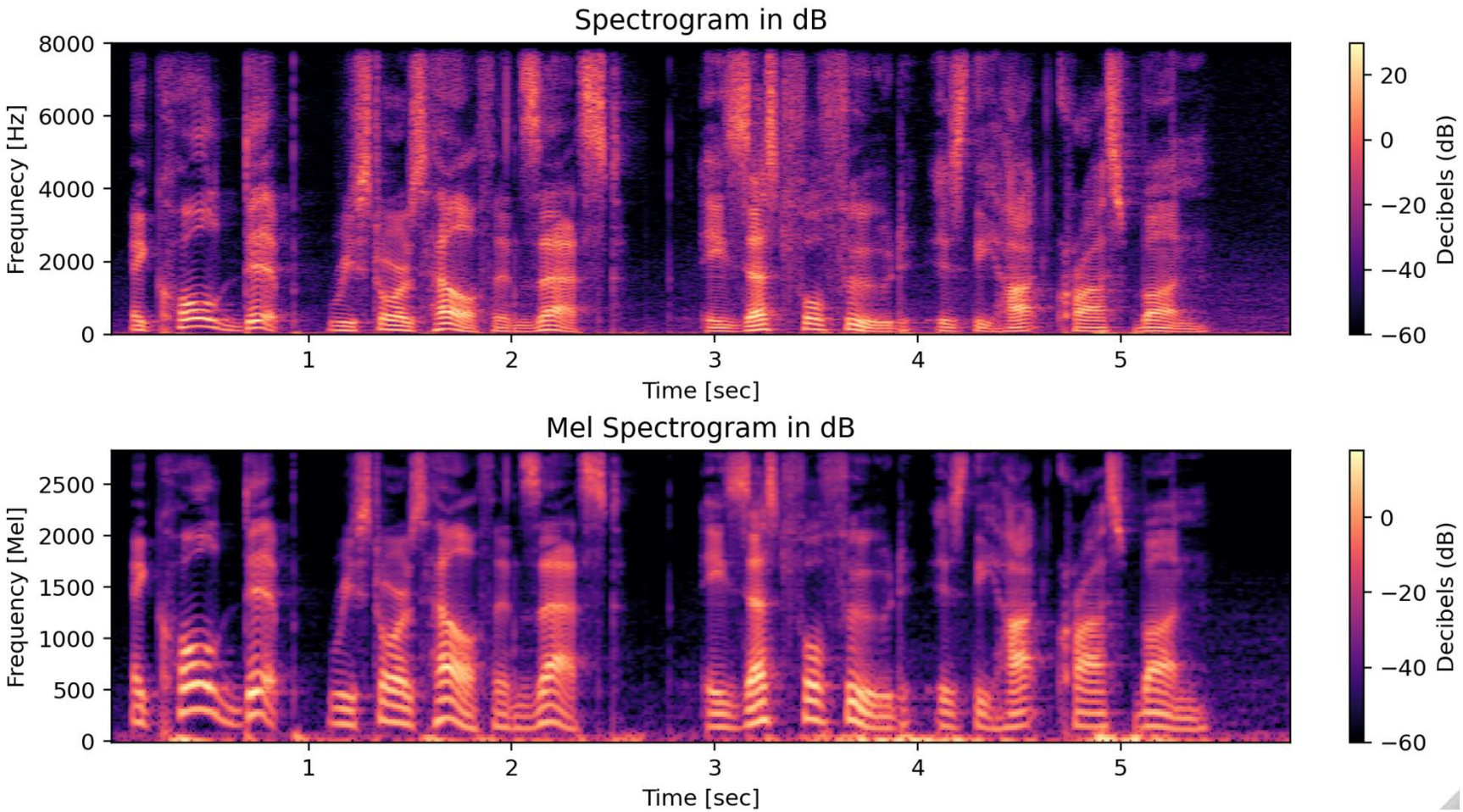
USE QUALITATIVE INTUITION, NOT ONE-TO-ONE PHYSICAL LABELS

35–55 MIN

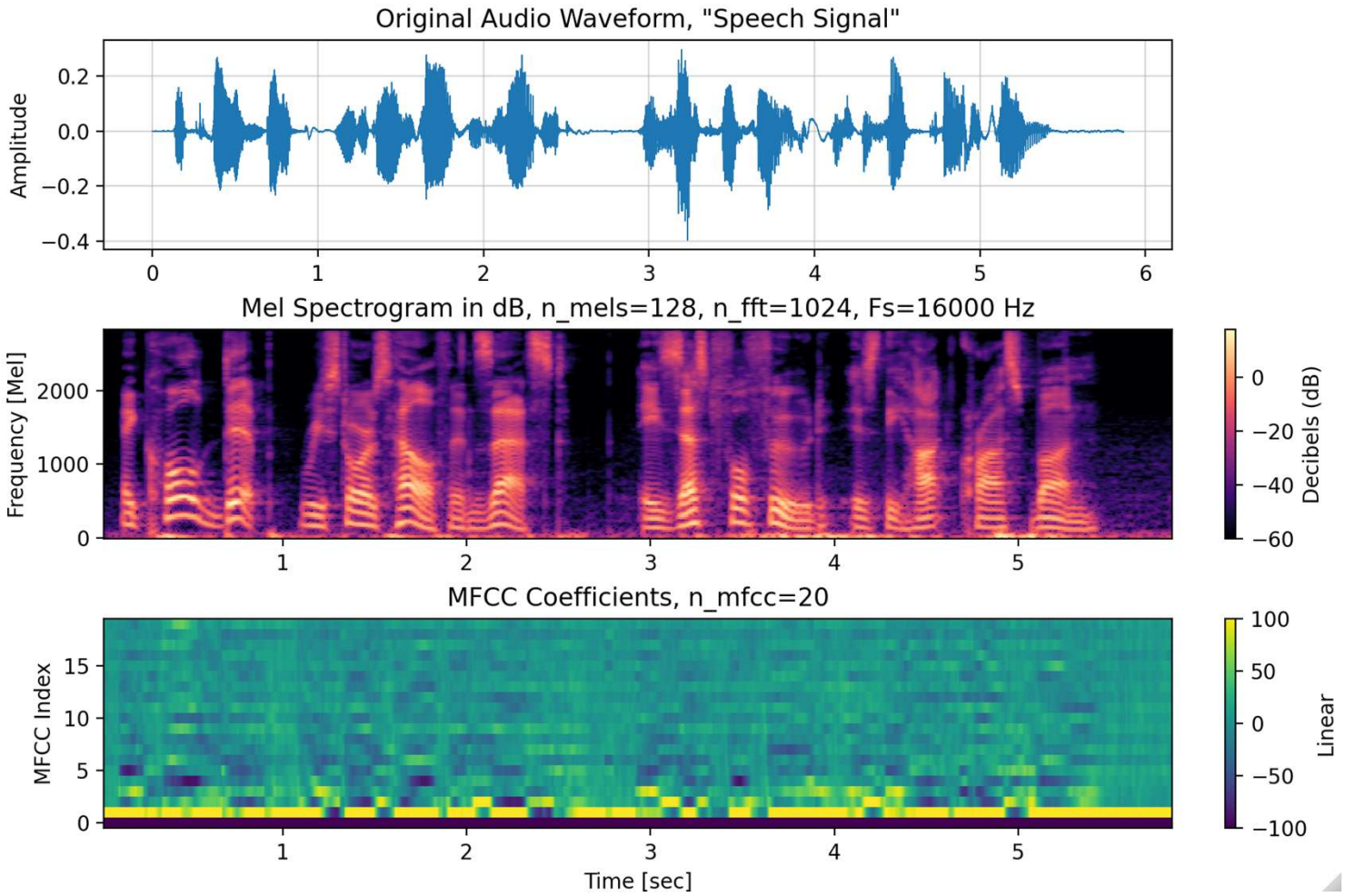


- Low-order coefficients: broad spectral-envelope / tilt information.
- Higher-order coefficients: progressively finer variation across Mel bands.
- MFCCs intentionally discard much detailed harmonic structure.
- That compression can help a classical classifier—or hurt a task needing fine time–frequency detail.

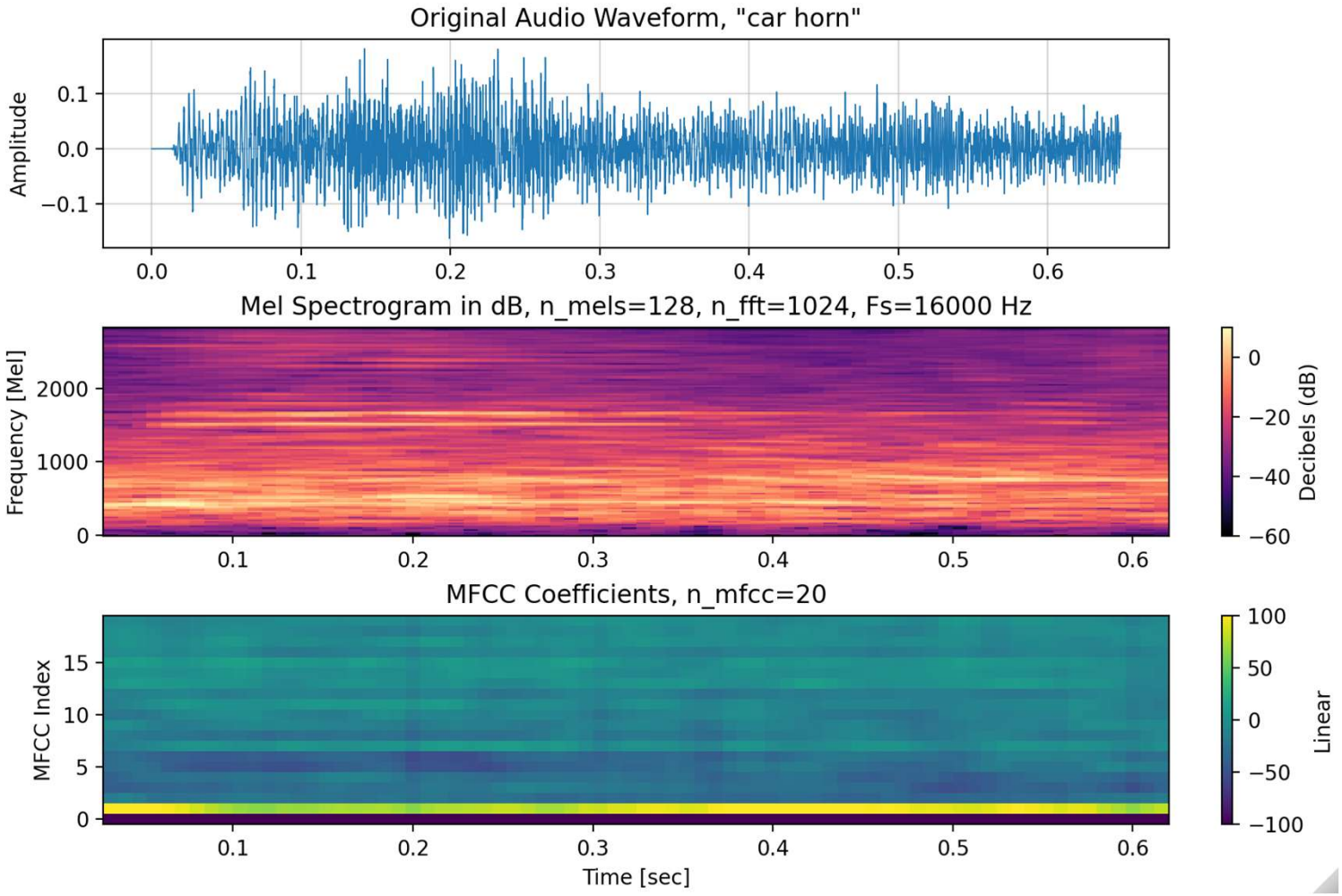
STFT Spectrogram in dB Compared to Mel Spectrogram in dB



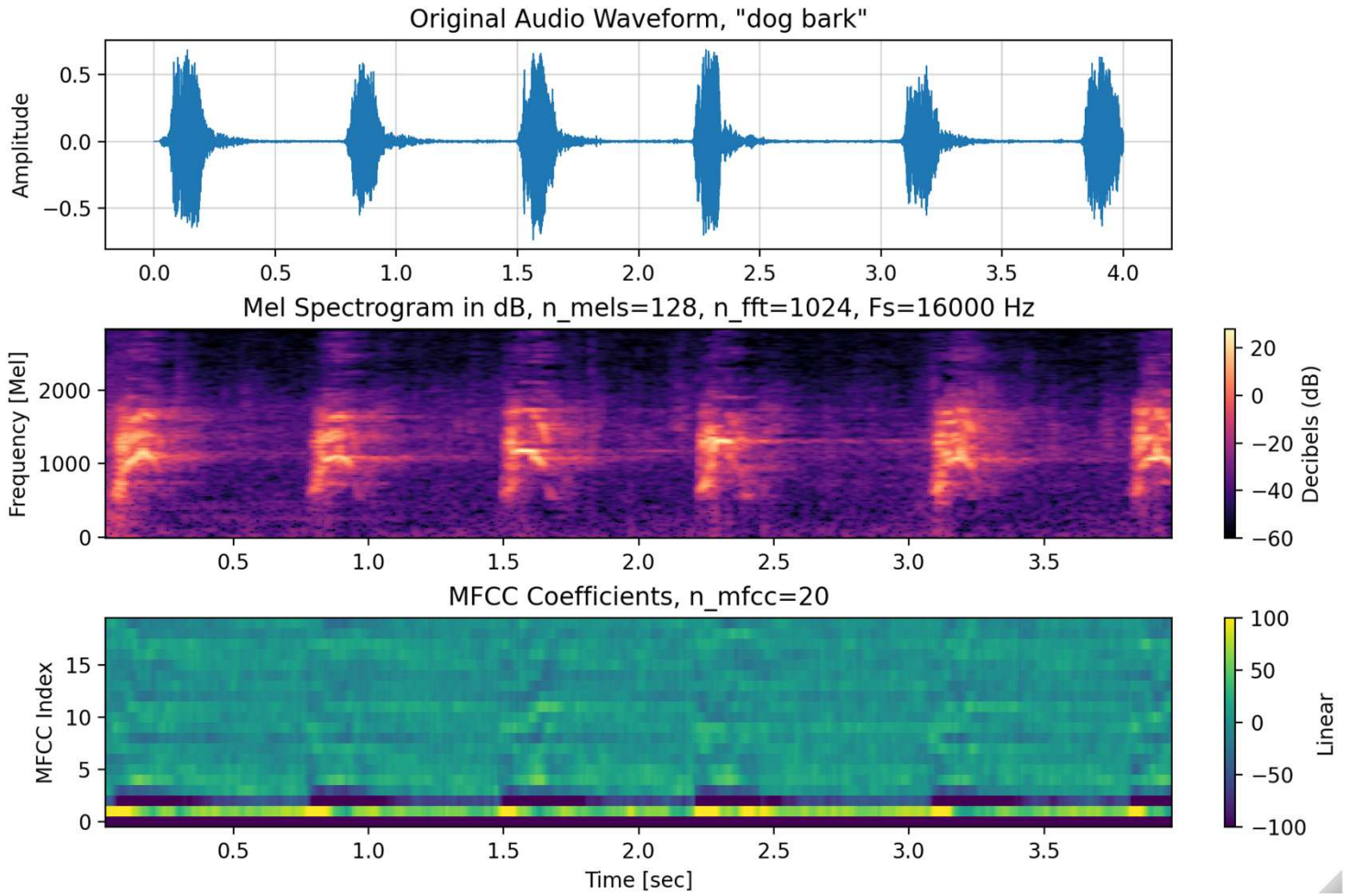
Log Mel Spectrogram and MFCCs for “Speech Signal”



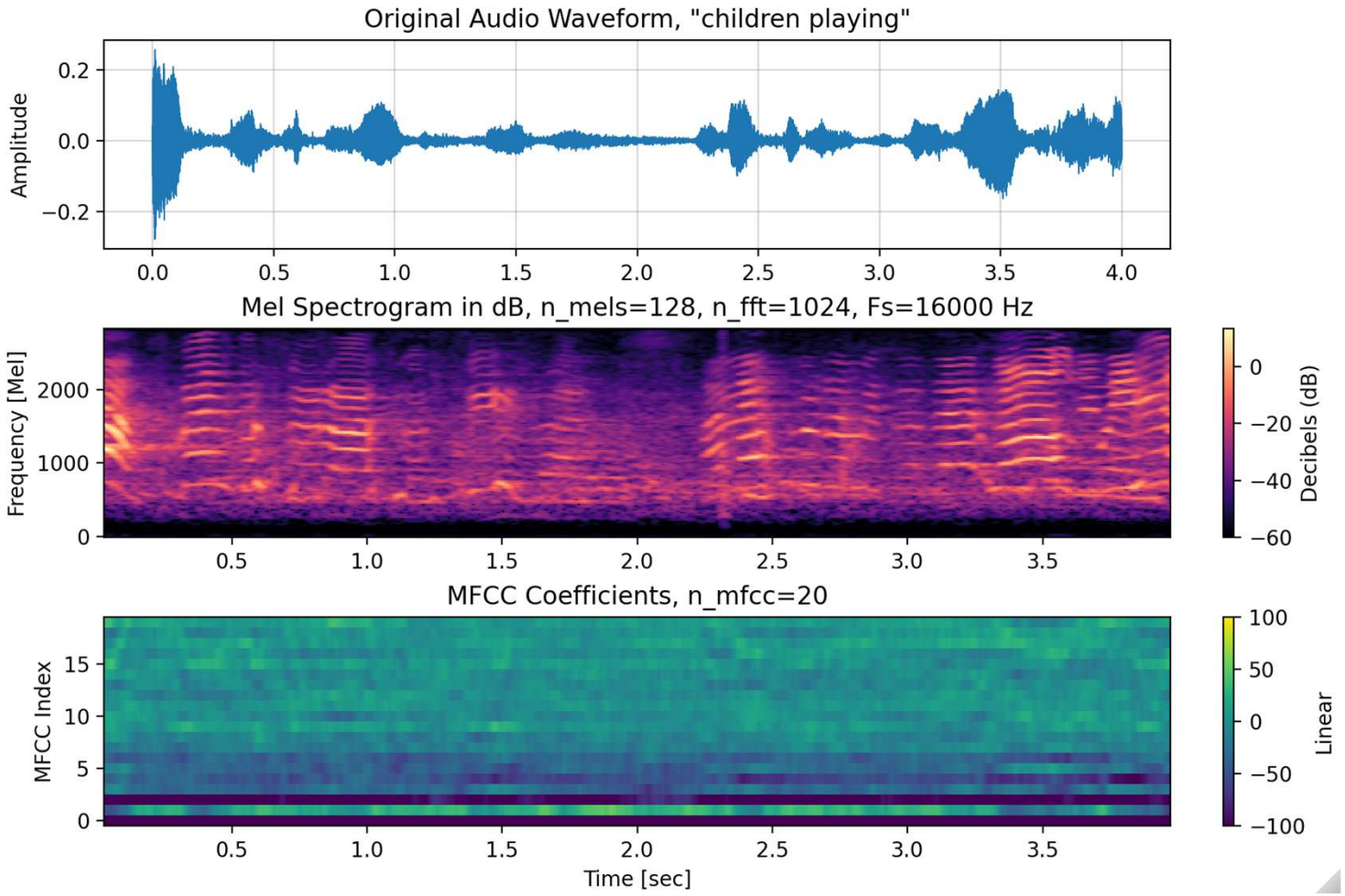
Log Mel Spectrogram and MFCCs for “Car Horn”



Log Mel Spectrogram and MFCCs for “Dog Bark”



Log Mel Spectrogram and MFCCs for “Children Playing”



Mel-spectrogram or MFCC?

CHOOSE THE REPRESENTATION TO MATCH THE MODEL AND TASK

35–55 MIN

Mel-spectrogram
2D time–frequency geometry
Natural CNN input
More local detail

MFCC matrix
Compact spectral-envelope view
Still has time frames
Classical baseline friendly

MFCC summary
Mean/std over time
Fixed vector
Temporal order destroyed

No universally “best” feature: representation creates an inductive bias.

Standardizing waveform geometry

BEFORE FEATURES, MAKE EXAMPLES COMPARABLE

55–75 MIN

1

Channels
Stereo → mono or a defined multi-channel policy

2

Sample rate
Resample every clip to 22.05 kHz

3

Duration
Pad/truncate to exactly 4.0 s

4

Amplitude
Use a consistent numeric range; do not erase meaningful level without justification

5

Features
Same n_fft, hop, n_mels, and n_mfcc for every clip

Downmixing stereo to mono

A CHANNEL DECISION IS PART OF THE MEASUREMENT MODEL

55–75 MIN

$$y_mono[n] = 0.5(y_L[n] + y_R[n])$$

- Simple averaging is common when spatial information is irrelevant.
- Do not downmix blindly if direction or spatial cues are part of the label.
- librosa.load(..., mono=True) can downmix automatically.

Feature engineering begins with deciding what information you are willing to throw away.

Resampling: one physical frequency grid for every example

22.05 KHZ BECOMES THE COURSE CONVENTION

55–75 MIN

$$N_{samples} = duration \times f_s = 4.0 \times 22050 = 88{,}200$$

- A fixed sample rate makes FFT-bin frequencies and Mel bands comparable across files.
- Always use an anti-aliased resampler; naive sample dropping is not resampling.
- 22.05 kHz implies Nyquist = 11.025 kHz—acceptable only if the task does not need higher-frequency information.

DSP prerequisite
Week 2 still matters: bad rate conversion can create aliases.

Padding and truncation: force a fixed tensor shape

SAME DURATION → SAME NUMBER OF FRAMES

55–75 MIN

```
target_len = int(4.0 * 22050)   # 88,200 samples

            if len(y) < target_len:
        y = np.pad(y, (0, target_len - len(y)))
                    else:
                y = y[:target_len]

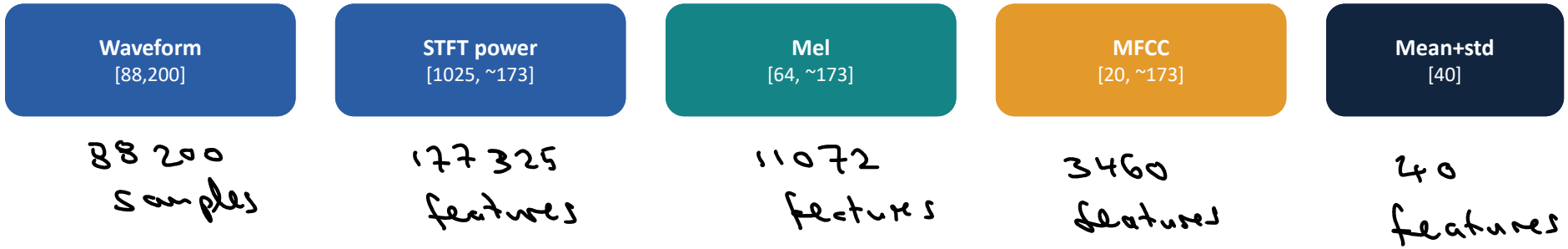
          assert y.shape == (target_len,)
```

- Padding location is a design choice: right, left, or centered.
- Truncation can remove events—random crops or multiple crops may be better during training.
- For this week, deterministic right-padding/right-truncation keeps the baseline easy to debug.

Shape bookkeeping: predict dimensions before running code

A POWERFUL DEBUGGING HABIT

55–75 MIN



- Exact frame count depends on STFT centering/padding conventions.
- Flattening 20×173 MFCCs gives 3460 features; mean+std gives only 40.
- Dimensionality reduction is useful—but every reduction throws away structure.

Live code-along: one UrbanSound8K clip

MAKE EVERY PREPROCESSING DECISION VISIBLE

55–75 MIN

```
y, sr0 = librosa.load(path, sr=None, mono=True)
y = librosa.resample(y, orig_sr=sr0, target_sr=22050)
y = fix_length(y, int(4.0 * 22050))

mel = librosa.feature.melspectrogram(
y=y, sr=22050, n_fft=2048,
hop_length=512, n_mels=64, power=2.0)
mel_db = librosa.power_to_db(mel, ref=np.max)
mfcc = librosa.feature.mfcc(S=mel_db, n_mfcc=20)
```

Instructor cue: Pause after every line: what invariant did we establish, and what information might we have discarded?

Concept check

REASON ABOUT REPRESENTATION CHOICES

CHECK

- A. Increasing n_mels always improves classification accuracy.
- B. Converting stereo to mono is harmless for every audio task.
- C. Mean MFCCs reduce dimensionality but remove temporal ordering.
- D. Converting power to dB restores information lost by Mel pooling.

Concept check

REASON ABOUT REPRESENTATION CHOICES

CHECK

- A. Increasing n_mels always improves classification accuracy.
- B. Converting stereo to mono is harmless for every audio task.
- C. Mean MFCCs reduce dimensionality but remove temporal ordering.
- D. Converting power to dB restores information lost by Mel pooling.

Correct: C

- A, B, and D are not guaranteed. Each preprocessing choice encodes an assumption about which information matters.

Lecture 25 synthesis

FROM WAVEFORM PHYSICS TO ML-READY FEATURES



- Mel features compress frequency according to a perceptual warping.
- MFCCs further rotate/compress log-Mel information for compact classical features.
- Fixed channels, rate, duration, and feature parameters give the model consistent geometry.
- Next lecture: feature matrices become X, labels become y, and evaluation discipline becomes central.